This chapter is the revision question bank for the AFCS (Automatic Flight Control System) section — Chapters 25–30. All questions with worked answers and brief explanations are provided below.

Quick Answer Keys

Paper 1

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Paper 2

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Paper 1 — Questions & Explanations

1. A single axis autopilot system:

a. provides stabilization about the normal axis

b. provides control about the pitch axis

c. is unsuitable for use in powered aircraft

d. provides control about the roll axis ✓

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A single-axis autopilot = wing leveller = roll axis control only.

2. A single axis autopilot may also be called:

a. altitude hold

b. wing leveller ✓

c. pitch control loop

d. auto-stabilization loop

3. An autopilot:

a. is a system which will maintain a preselected altitude

b. is a system which will maintain a preselected airspeed

c. is an auto-stabilization system ✓

d. is an outer loop control system

> The fundamental definition: an autopilot is an auto-stabilization system — it maintains a stable reference condition. Altitude/speed hold are specific modes, not the definition.

4. The fundamental components of an autopilot control loop are:

a. rate gyro, servomotor, error signal generator ✓

b. rate gyro, servomotor, torque limiter

c. torque limiter, error signal generator, servomotor

d. servomotor, rate gyro, torque limiter, error signal generator

5. A device in a closed loop control system in which a small power input controls a much larger power output in a strictly proportionate manner is:

a. an amplifier

b. a servomechanism ✓

c. a powered flying control unit

d. a rate gyro

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6. An automatic flight control system:

a. is another name for an autopilot system ✓

b. applies flight data to the autopilot system

c. is automatically disengaged by a GPWS alert

d. can only be used in EFIS equipped aircraft

7. An aircraft has yaw damping included in its auto-stabilization system. An essential requirement of such a system is:

a. a three axis autopilot system ✓

b. parallel connected servomotors

c. automatic maintenance of C of G position

d. INS inputs to the CADC

> Yaw damping is a 3rd-axis (yaw) function — so a 3-axis autopilot is the essential requirement.

8. Automatic flight systems may be capable of controlling the aircraft flight in:

a. azimuth, elevation and velocity ✓

b. azimuth and velocity only

c. azimuth only

d. azimuth and elevation only

9. An automatic flight control system is fitted with control wheel steering (CWS):

a. The autopilot must be disengaged before the pilot can input manoeuvring commands

b. Manoeuvring commands may be input by applying normal forces to the control yoke without first disengaging the autopilot ✓

c. Commands via pitch and turn controls on the AFS panel without disengaging

d. The CWS is only there for steering on the ground

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10. During an approach to an autoland at 1500 feet:

a. off line channels are manually engaged, flare mode is armed

b. localizer is controlling the roll channel, off line channels are automatically engaged and flare mode is armed ✓

c. localizer controlling roll, stabilizer trimmed nose up and roll out armed

d. provided both LOC and G/S valid LAND 3 will illuminate

11. What type of autoland system would be required for the landing to continue following a single failure below alert height?

a. Fail-soft

b. Fail-passive

c. Fail-operational or fail-active ✓

d. Land 2 system

> Fail-OPERATIONAL = landing continues after single failure below alert height (LAND 3). Fail-PASSIVE (LAND 2) = no unsafe deviation but crew must go-around.

12. Inputs to the rudder channels initially originate from:

a. servomotors

b. compass gyro and gyro for AH

c. compass gyro and turn and slip gyro ✓

d. AH gyro and turn and slip gyro

13. An automatic flight system which can safely continue with an automatic landing after a system failure is a:

a. fail-redundant system

b. fail-passive system

c. three axis system

d. fail-operational system ✓

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14. Altitude Select and Altitude Hold are examples of:

a. inner loop functions in pitch

b. outer loop functions in pitch (but displayed as "roll" in some texts — here the answer is b: outer loop functions) ✓

c. interlocking functions

d. outer loop functions in roll

> ALT SEL and ALT HOLD are outer loop pitch channel functions. The answer key gives **b**. Note: the key says "b" which is labelled "manometric functions from the ADC" in some versions — but the standard answer is outer loop pitch = (b) as given in the answer key.

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15. During an autoland the caption LAND 2 is illuminated. The system is:

a. fail-active or fail-operational

b. fail-passive ✓

c. approaching decision height

d. requiring a crew input

16. For an autoland system to meet FAIL-PASSIVE criteria it must:

a. have suitable system redundancy

b. withstand a system failure without excessive deviations from flight path ✓

c. can continue with an autoland below alert height

d. can continue with an autoland above alert height

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17. During an autoland at 50 ft AGL (45' GA) the pitch control is _____ and roll control is _____:

a. glide slope / localizer

b. glide slope / roll out

c. flare / roll out

d. flare / localizer ✓

> At 50 ft: FLARE mode controls pitch; LOCALIZER still controls roll (roll-out arms at 50 ft but engages at lower height). At 27 ft roll-out engages.

18. During an autoland approach:

a. flare is engaged at 1500' AGL

b. localizer roll control is disengaged just prior to touchdown

c. flare is disengaged prior to touchdown at 5' GA ✓

d. glide slope is the engaged pitch mode until 5' GA

19. In an autoland at 1000' AGL with two autopilots engaged:

a. the armed roll mode would be LOCALIZER

b. the engaged roll mode would be GLIDE SLOPE

c. the engaged pitch mode would be FLARE

d. the engaged roll mode would be LOCALIZER ✓

20. An automatic flight control system in which applying normal forces on the control column allows the pilot to input demands to the autopilot is a:

a. control wheel steering ✓

b. touch control steering

c. series connected system

d. parallel connected system

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21. If a fault develops in a triplex autopilot system during an approach, the system will revert to:

a. fail-passive and the landing may continue ✓

b. fail control wheel mode

c. fail-operational

d. a manual disconnect

> Triplex → one failure → duplex (fail-passive/LAND 2). The landing can still continue but must be monitored for further failures.

22. CADCs transmit data concerning:

a. airspeed, altitude and decision height

b. airspeed, altitude and Mach number ✓

c. airspeed, attitude and Mach number

d. airspeed and altitude only

23. Inner loop stability is obtained by:

a. inputs from the ADC

b. manometric locks

c. rate gyro displacement ✓

d. raw data feed to the data control bus bar

24. The autothrottle is used to control some factors during the three primary control modes; they are:

a. EPR, Mach and Speed ✓

b. EPR, wheel and speed

c. EPR, Mach and altitude

d. EPR, wheel and altitude

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25. The mode that enables the pilot to manoeuvre in pitch and roll using the automatic control system is:

a. control wheel steering (CWS) mode — when the wheel is released the aircraft holds the newly established attitude ✓

b. touch control steering via ADC

c. CWS mode which disengages servomotors

d. TCS mode which prevents flap retraction

26. Touch control steering:

a. prevents aerodynamic feedback

b. operates only with flaps down

c. allows the pilot to control the aircraft with the servomotors disengaged ✓

d. engages servomotors during manual pitch and roll

27. A system which can still function without degradation of performance after a failure has:

a. fail-passive ability

b. fail-soft ability

c. fail-operational ability ✓

d. fail-symbol ability

28. During a CAT 2 ILS automatic approach, the source for altitude information is the:

a. basic altitude capsule stack

b. radar altimeter effective below ~2500 feet

c. radio altimeter effective below ~2500 feet ✓

d. mode comparator sensor

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29. Heading hold mode relates to control in:

a. height lock via CADC

b. pitch channel via inner loop

c. roll channel via the outer loop control source ✓

d. manometer mode of CADC

30. The system which allows the pilot to control the aircraft with the servomotors engaged is called:

a. touch control steering

b. control wheel steering ✓

c. electronic inner/outer axis loop

d. outer loop control

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31. The type of automatic landing system which would necessitate a manual landing after a system failure is:

a. fail-passive ✓

b. fail-safe

c. fail-active

d. fail-operational

32. After a failure of one of the necessary redundant systems below alert height you would:

a. continue the descent but revert to a higher DH

b. carry out a missed approach

c. disengage autoland and take over manually

d. continue descent and land automatically ✓

> Below alert height in a FAIL-OPERATIONAL system: one failure → still fail-passive → continue autoland. This is the definition of fail-operational.

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33. When localizer and glide slope are captured at 1500 feet during an automatic landing sequence, two other functions will also activate; they are:

a. touchdown mode and roll out mode

b. flare mode arm and touchdown mode

c. flare mode engage and roll out mode

d. flare mode arm and off line channels engaged ✓

34. A fundamental requirement of a closed loop servomechanism is:

a. a stable reference device

b. an interlock control

c. a tacho-generator

d. feedback ✓

35. ALT HOLD is an example of:

a. inner loop control in the roll axis

b. outer loop input to the pitch channel ✓

c. outer loop control about the longitudinal axis

d. inner loop control in the pitch axis

36. A rate gyro:

a. has three degrees of freedom, two gimbals and a transducer

b. senses rate of turn and positions an indicator on the EHSI

c. supplies rate and displacement information to the computer ✓

d. controls the outer loop inputs

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37. To prevent servomotor runaway from producing excessive demands to the control surface:

a. a gyro damper is fitted

b. a torque limiter is fitted ✓

c. a gyro limiter is fitted

d. a torque converter is fitted

38. Auto-trim is functional:

a. in the pitch and roll channel with the autopilot engaged

b. in the pitch channel only with the autopilot engaged ✓

c. in the pitch channel only with the autopilot disengaged

d. in the pitch and roll channel with the autopilot disengaged

39. L NAV is an ___ input to the ___ channel using data from the ___:

a. outer loop, pitch, FMC

b. inner loop, pitch, ADC

c. outer loop, roll, FMC ✓

d. inner loop, roll, ADC

40. In an aircraft which requires a Mach trim system it will apply inputs to the horizontal stabilizer:

a. all the time

b. at high Mach numbers with the autopilot engaged

c. at Mach one with the autopilot engaged or not

d. at high subsonic speeds with the autopilot engaged or not ✓

Paper 2 — Questions & Explanations

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1. With autopilot in Alt mode the Captain alters the barometric setting. The aircraft:

a. maintains its altitude ✓

b. changes altitude per pressure setting

c. switches barometric input to 1st Pilot setting

d. trips out of altitude hold

> ALT HOLD maintains the actual altitude; changing the barosetting changes what altitude is *displayed*, not what is held.

2. Control wheel steering enables a pilot to:

a. taxi the aircraft on the ground

b. manoeuvre the aircraft in the air while the autopilot is engaged ✓

c. alter the flight path while the autopilot is engaged by applying a breakout force

d. manoeuvre the aircraft with the autopilot disengaged

3. Autopilot synchronization in an aircraft:

a. requires interlocks to be made before AP engages

b. ensures that when the autopilot is engaged, the take-over is smooth and without snatching ✓

c. requires the aircraft to be trimmed before AP can engage

d. needs at least two alternators in parallel

4. The rules for the use of auto-trim are that it:

a. can be engaged without the autopilot

b. usually operates on all three axes

c. is not needed if the autopilot is engaged

d. operates only in conjunction with the autopilot ✓

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5. JAR OPS requirements for single pilot IFR state the aircraft must be fitted with:

a. a single axis autopilot

b. a two axis autopilot ✓

c. a three axis autopilot

d. a two axis autopilot with autothrottle

6. JAR 25 autopilot installation requirements — A: automatic synchronization. B: quick release controls on both control wheels:

a. Only A correct

b. Both statements are correct ✓

c. Only B correct

d. Neither correct

7. Flight envelope protection: A = High speed protection prevents exceeding VMO/MMO. B = High AoA protection comes in when aircraft reaches stalling AoA:

a. Only A correct

b. Only B correct

c. Both correct

d. Neither statement is correct ✓

> Neither is fully correct: High speed protection prevents reaching Vd/Md (not just VMO/MMO). High AoA protection activates at α-floor, *before* the stalling AoA is reached.

8. The control laws for an autopilot are known as:

a. normal law and emergency law

b. alternate law and direct law

c. normal, alternate and emergency laws

d. normal, alternate and direct laws ✓

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9. An autoland system that continues to function without degradation of performance after an AP failure would be:

a. fail-passive

b. fail-safe

c. fail-operational ✓

d. duplex

10. The autoland sequence is considered complete when:

a. reverse thrust is engaged

b. the autopilot is manually disengaged by the pilot ✓

c. the aircraft touches down

d. the aircraft reaches the end of the runway

11. The autothrottle will come on automatically even with A/T switch OFF when:

a. in an FBW aircraft the AoA reaches a critical value called α-floor ✓

b. the AoA reaches the stalling angle

c. TOGA button is pressed

d. reverse thrust is selected in flight

12. An aircraft on autopilot in VOR mode loses VOR signals through the cone of silence. The autopilot:

a. automatically switches to Heading mode

b. decouples from VOR and disconnects

c. tunes to the next VOR on the route

d. decouples from the VOR and flies the last heading for a fixed period ✓

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13. For a non-synchronized autopilot, snatching when engaging can be prevented by:

a. the pilot ensuring the aircraft is trimmed out before selecting or disengaging the autopilot ✓

b. being in straight and level

c. disengaging the auto-trim

d. switching on the yaw dampers

14. With autopilot in CWS the pilot manoeuvres the aircraft and releases control. The aircraft will maintain:

a. heading and altitude

b. heading, speed and attitude

c. altitude and attitude

d. attitude at the time of release ✓

15. Autopilot corrections affecting pitch are carried out by:

a. auto-trim only

b. auto-trim and elevators ✓

c. elevators only

d. autothrottle

16. For commercial aircraft single pilot IFR, the minimum autopilot control requirement is:

a. three axes

b. heading mode

c. altitude hold and heading mode ✓

d. altitude hold, heading mode and speed

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